AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH

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Assignment Title:	Personal Finance Tracker		
Assignment No:	01	Date of Submission:	28 March 2026
Course Title:	Programming In Python		
Course Code:	CSC4162	Section:	A
Semester:	Spring 2025-26	Course Teacher:	MD. TANZEEM RAHAT

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FACULTY COMMENTS	**Marks Obtained**	
	Total Marks	

PROJECT REPORT

Personal Finance Tracker (CLI Application)

1. Project Overview

The Personal Finance Tracker is a command-line based application developed using Python. It helps users manage income and expenses efficiently through a menu-driven system.

2. Track Name

Command Line Interface – CLI Project

3. Technology Stack

- Language: Python 3
- Interface: CLI
- Storage: JSON
- Library: rich (for enhanced terminal output)

4. Setup & Run Instructions

Create virtual environment:

python -m venv .venv

Activate environment:

Windows: .venv\Scripts\activate

macOS/Linux: source .venv/bin/activate

Install dependencies:

pip install rich

Generate requirements file:

pip freeze > requirements.txt

Run the application:

python main.py

5. Project Structure

- finance_cli/
- cli.py # CLI interface
- manager.py # Business logic
- models.py # Data model
- storage.py # JSON handling

6. OOP Design

- models.py: Represents transaction entity
- manager.py: Handles CRUD, search, filter, reports
- storage.py: Manages JSON data storage
- cli.py: Controls user interaction
- Uses encapsulation, modularity, and composition

7. Features

- Add, view, update, delete transactions
- Search and filter transactions
- Monthly report generation
- Category breakdown
- Dashboard showing income, expense, balance
- Automatic data saving

8. Data Persistence

- Data stored in JSON file
- Loaded at program start
- Saved after every operation
- No external database required

9. Validation & Error Handling

- Handles invalid input
- Ensures numeric values
- Prevents empty fields
- Shows user-friendly messages
- Prevents crashes

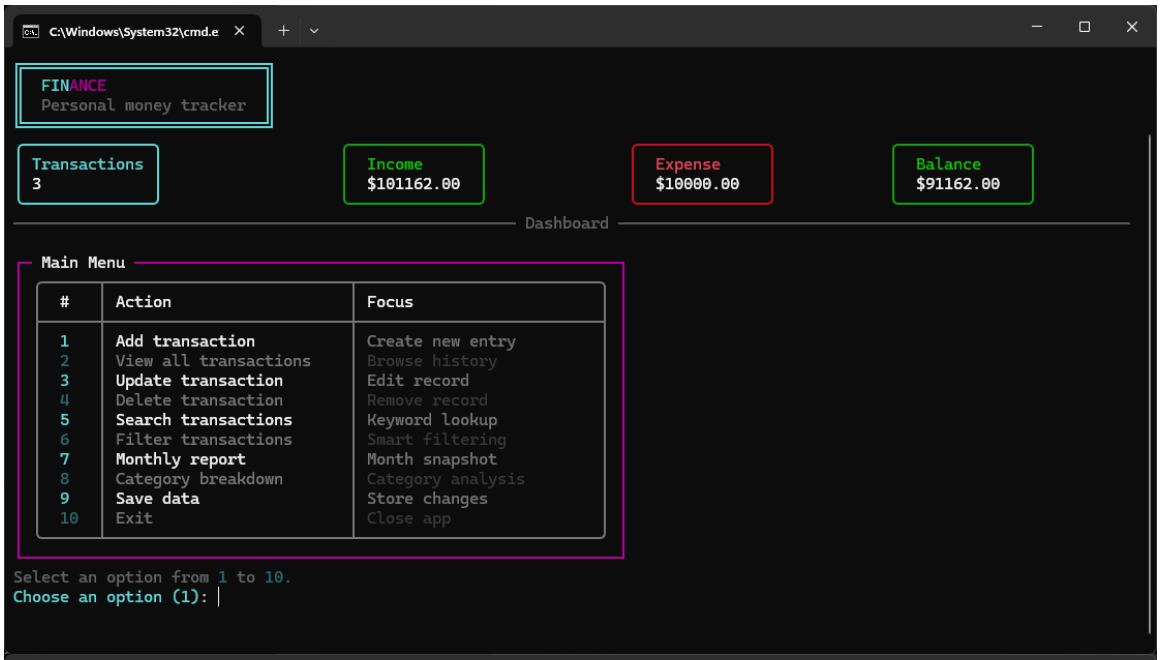
10. Demo Notes

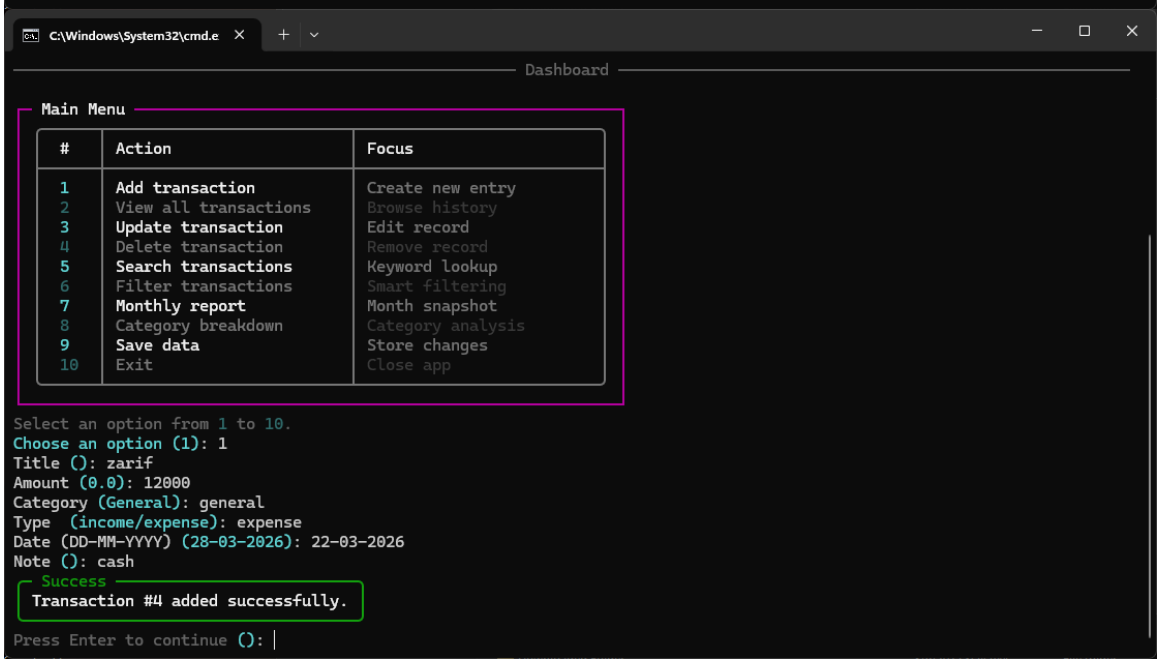
- Menu-driven CLI system
- User selects options and inputs data
- System processes and displays results instantly

11. Conclusion

A lightweight and efficient financial tracking system using Python CLI Demonstrates OOP, file handling, and real-world application design

12. Screenshots





C:\Windows\System32\cmd.e

#	Action	Focus
1	Add transaction	Create new entry
2	View all transactions	Browse history
3	Update transaction	Edit record
4	Delete transaction	Remove record
5	Search transactions	Keyword lookup
6	Filter transactions	Smart filtering
7	Monthly report	Month snapshot
8	Category breakdown	Category analysis
9	Save data	Store changes
10	Exit	Close app

```
Select an option from 1 to 10.
Choose an option (1): 2
Sort by [date/amount/category/type/title] (date): amount
Sort descending? [y/n] (y): y
```

Transaction History

ID	Date	Type	Category	Title	Amount	Note
2	12-12-2012	Income	Local	hulu	$100131.00	no need
4	22-03-2026	Expense	General	zarif	$12000.00	cash
3	20-01-2001	Expense	Local	arshad	$10000.00	gpt
1	20-12-2014	Income	Local	maruf	$1031.00	cash

```
Press Enter to continue (): |
```

C:\Windows\System32\cmd.e

#	Action	Focus
1	Add transaction	Create new entry
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10	Exit	Close app

```
Select an option from 1 to 10.
Choose an option (1): 7
Enter year (2026): 2026
Enter month number (3): 1
```

Monthly Report 2026-01

Metric	Value
Transactions	0
Total Income	$0.00
Total Expense	$0.00
Net Balance	$0.00

```
Press Enter to continue (): |
```

C:\Windows\System32\cmd.e

Dashboard

Main Menu

#	Action	Focus
1	Add transaction	Create new entry
2	View all transactions	Browse history
3	Update transaction	Edit record
4	Delete transaction	Remove record
5	Search transactions	Keyword lookup
6	Filter transactions	Smart filtering
7	Monthly report	Month snapshot
8	Category breakdown	Category analysis
9	Save data	Store changes
10	Exit	Close app

```
Select an option from 1 to 10.
Choose an option (1): 8
Show category breakdown for [expense/income] (expense): income
```

Income Category Breakdown

Category	Total	Share
Local	$101162.00	100.0%

```
Press Enter to continue (): |
```

C:\Windows\System32\cmd.e

Personal money tracker

Transactions
4

Income
$101162.00

Expense
$22000.00

Balance
$79162.00

Dashboard

Main Menu

#	Action	Focus
1	Add transaction	Create new entry
2	View all transactions	Browse history
3	Update transaction	Edit record
4	Delete transaction	Remove record
5	Search transactions	Keyword lookup
6	Filter transactions	Smart filtering
7	Monthly report	Month snapshot
8	Category breakdown	Category analysis
9	Save data	Store changes
10	Exit	Close app

```
Select an option from 1 to 10.
Choose an option (1): 9
```

Success

Data saved successfully.

```
Press Enter to continue (): |
```

```
C:\Windows\System32\cmd.e

Personal money tracker

Transactions        Income          Expense         Balance
4                   $101162.00      $22000.00       $79162.00

                    Dashboard

Main Menu

| # | Action                | Focus                |
|---|---|---|
| 1 | Add transaction       | Create new entry     |
| 2 | View all transactions | Browse history       |
| 3 | Update transaction    | Edit record          |
| 4 | Delete transaction    | Remove record        |
| 5 | Search transactions   | Keyword lookup       |
| 6 | Filter transactions   | Smart filtering      |
| 7 | Monthly report        | Month snapshot       |
| 8 | Category breakdown    | Category analysis    |
| 9 | Save data             | Store changes        |
| 10 | Exit                 | Close app            |

Select an option from 1 to 10.
Choose an option (1): 10

Thank you for using Personal Finance CLI.

C:\Users\ZARIF-ROG\Desktop\Python project>
```